

KUWAIT INTL, Kuwait

WMO: 405820

Lat: 29.227N

Lon: 47.969E

Elev: 206

StdP: 14.59

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.4	42.7	12.6	10.5	68.2	17.2	13.2	73.4	25.0	62.2	22.8	61.6	4.8	300	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	118.6	70.1	116.8	69.6	115.0	69.2	83.1	95.2	80.0	95.0	77.1	95.1	14.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	158.2	91.7	76.1	137.2	89.6	71.8	118.4	87.3	47.4	94.5	44.0	94.9	40.8	95.0	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	22.4	20.4	DB	34.4	122.0	3.2	1.3	32.1	122.9	30.3	123.6	28.5	124.3	26.1	125.2	(4)
(5)			WB	30.0	86.4	3.9	2.5	27.2	88.2	24.9	89.7	22.7	91.1	19.9	92.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.6	56.5	60.4	68.8	79.4	91.2	99.2	102.0	101.2	94.6	84.2	69.1	59.2		
	DBStd	17.26	5.22	5.60	5.75	5.86	4.95	3.64	3.33	3.33	4.31	4.98	6.67	5.65		
	HDD50	13	8	2	0	0	0	0	0	0	0	0	0	3		
	HDD65	668	267	147	26	1	0	0	0	0	0	0	33	194		
	CDD50	11178	210	292	583	881	1278	1477	1611	1588	1337	1060	574	287		
	CDD65	6357	4	17	144	432	813	1027	1146	1123	887	595	156	13		
	CDH74	105291	14	110	1316	5133	13254	18855	21506	20652	15082	7913	1375	81		
	CDH80	76034	2	18	434	2599	9105	14550	17043	16190	10936	4692	460	5		
(14) Wind	WSAvg	9.1	7.8	8.7	9.0	9.4	9.4	12.1	11.8	10.1	8.9	7.5	7.5	7.5		
(15) Precipitation	PrecAvg	4.4	0.9	0.4	0.6	0.4	0.1	0.0	0.0	0.0	0.0	0.1	0.7	0.8		
	PrecMax	8.2	3.7	2.8	2.6	1.7	1.1	0.0	0.0	0.0	0.6	1.5	4.6	2.6		
	PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.8	0.9	0.6	0.7	0.4	0.2	0.0	0.0	0.0	0.1	0.3	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	82.2	96.3	104.0	113.4	120.0	121.2	120.4	116.7	107.6	93.5	79.9		
		MCWB	56.5	58.1	61.3	65.2	67.8	69.9	70.3	70.4	69.9	67.1	63.4	58.3		
	2%	DB	71.5	77.0	89.2	98.7	110.8	116.9	118.6	118.4	113.3	104.0	89.2	76.5		
		MCWB	56.6	56.0	60.2	64.7	67.6	68.8	70.1	70.3	69.1	66.6	64.1	59.0		
	5%	DB	69.0	73.8	84.3	95.3	107.8	114.8	116.8	116.6	111.1	100.8	85.1	73.2		
		MCWB	56.1	55.7	58.9	63.8	66.7	67.9	69.5	70.1	68.5	66.3	63.2	58.8		
	10%	DB	66.5	71.5	80.8	91.8	105.4	112.9	114.9	114.7	108.7	98.4	82.0	69.9		
		MCWB	55.4	55.4	57.8	62.5	66.3	67.5	69.1	70.0	67.8	65.8	62.9	58.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.0	65.1	68.4	71.8	76.4	79.0	83.7	87.2	84.5	80.5	75.1	67.8		
		MCDB	68.3	70.7	76.8	85.1	90.4	96.9	97.2	96.4	92.8	87.1	81.1	71.6		
	2%	WB	62.3	63.2	66.2	69.8	73.2	73.7	79.9	84.5	81.3	78.4	72.5	65.7		
		MCDB	66.7	69.0	74.8	84.0	91.7	102.5	100.0	96.1	92.2	86.8	79.3	69.8		
	5%	WB	59.9	61.1	64.4	68.1	70.9	71.1	76.0	81.1	77.7	75.9	69.7	63.2		
		MCDB	64.9	67.4	73.9	83.7	94.1	107.9	102.9	97.0	93.8	85.7	77.7	68.4		
	10%	WB	57.6	59.0	62.4	66.7	69.3	69.4	72.6	77.0	73.5	73.4	66.8	60.5		
		MCDB	63.8	66.1	73.9	83.4	96.3	107.7	108.1	101.1	97.7	84.9	75.8	67.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.7	22.7	23.0	25.0	26.7	26.7	27.7	28.2	26.4	21.3	19.4		
		MCDBR	21.3	23.8	28.2	27.3	28.8	29.3	28.3	29.4	30.2	30.3	25.9	23.9		
		MCWBR	11.2	11.0	11.1	9.7	9.7	9.2	9.2	10.1	10.7	12.4	12.8	12.0		
		MCDBR	15.9	16.8	19.0	20.5	23.4	28.2	26.9	26.1	26.3	23.9	19.3	15.4		
	5% WB	MCWBR	10.4	10.1	10.2	9.1	10.0	10.9	13.7	14.5	14.3	12.9	12.2	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.506	0.553	0.608	0.695	0.697	0.653	0.672	0.623	0.569	0.570	0.544	0.507		
	taud		1.827	1.733	1.618	1.515	1.517	1.563	1.558	1.648	1.741	1.741	1.784	1.836		
	Ebn at Noon		221	223	222	209	209	218	213	222	228	215	207	212		
	Edh at Noon		56	68	81	92	92	88	88	80	70	66	59	54		
(44) All-Sky Solar Radiation	RadAvg		1093	1362	1715	1871	2154	2402	2304	2190	1952	1552	1138	1006		
	RadStd		77	87	75	98	97	82	61	49	65	63	88	90		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+193	
(47)	Regional (11 neighbors)	+0.59	N/A	N/A	+0.97	N/A	N/A	N/A	N/A	+169	+193	(47)

Nomenclature: See separate page